Ecosystem Explorer — One-Page Summary

What it does

The Ecosystem Explorer simulates how reallocating developed urban land across three land use types affects flood runoff, urban cooling, and food production simultaneously. It is designed for comparative exploration of tradeoffs in early-stage planning conversations — not for precise prediction or final investment decisions.

Three land use types

Land Use	NLCD Code	Best for
Green Infrastructure	90 (woody wetlands)	Flood reduction
Food Forest	41 (deciduous forest proxy)	Cooling + food production
High Density	24 (developed high intensity)	— (worst for all three)

Five outcome metrics

Metric	What it measures	Unit
Flood Risk Reduction	Runoff potential derived from average Curve Number; higher = less runoff	index 0–100
Temperature Change	Approximate air-temperature difference vs unmodified baseline	°F vs baseline
Runoff Volume	Total stormwater volume for a 2-inch design storm across all developed land	acre-feet
Food Production	Annual yield from food forest pixels, assuming mature managed systems	M lbs/yr
Est. Implementation Cost	Order-of-magnitude capital cost based on per-acre cost sliders	$M

How it's computed

- **Flood and runoff** — USDA SCS Curve Number method applied per pixel using NLCD land cover and SSURGO soil hydrologic group; aggregated to city-wide mean CN and total runoff volume for a 2-inch design storm.
- **Cooling** — InVEST Urban Cooling Model HM index (`HM = (shade + kc) / 2` per pixel), converted to °F using a 4°F/HM unit factor (approximate midpoint from InVEST literature; ±2°F uncertainty).
- **Food production** — Pixel count of food forest conversions × 0.222 acres/pixel × 11,500 lbs/acre/year benchmark from NatCap food forest studies.

Key assumptions and limitations

- **City-scale aggregates only** — results are city-wide means; spatial placement of conversions is random (or intensity-weighted in heat-priority mode), not parcel-aware.
- **Temperature calibration not locally validated** — the 4°F/HM unit factor is a literature midpoint for Minneapolis; outputs carry roughly ±2°F uncertainty.
- **Food yield is a flat benchmark** — 11,500 lbs/acre/year assumes a mature, well-managed food forest; actual yield varies by species, soil, and management.
- **Implementation costs are illustrative** — default per-acre values ($50k GI, $10k FF, $5k HD) are order-of-magnitude placeholders, not sourced from specific projects.
- **Surrogate does not cover cost or heat-priority mode** — the optimizer searches ecological outcomes only; cost constraints and spatial weighting effects are not included in surrogate predictions.

Computation architecture

The full raster model provides the detailed spatial biophysical calculations. The lookup table makes the app responsive. The surrogate model makes large-scale scenario search possible.